Data Analytics

Project 2

Industrial Training Kit

Batch: 2026 | Powered by DecodeLabs

WELCOME TO THE TEAM! 🚀

Step into the role of a Data Analyst at DecodeLabs. Project 2 is your discovery phase: Exploratory Data Analysis (EDA). This track isn't about "simple reporting"—it's about Uncovering the Story. Before you build complex dashboards or predictive models, you must master the art of interrogating data to find hidden patterns, trends, and outliers. By completing this milestone, you are proving you can transform a static table of numbers into meaningful insights through pure analytical logic. Let’s design a digital foundation that turns raw information into absolute clarity.



GUIDELINES:

Qualification Criteria

EARN YOUR BADGE 🛡️

- Requirement: Complete project 2 to unlock other projects for next week.
- Standard: All projects must be verified for quality.
- Mandatory: Project 2 is the vital bridge to professional development for every intern.





Project 2: Exploratory Data Analysis (EDA)

Goal:

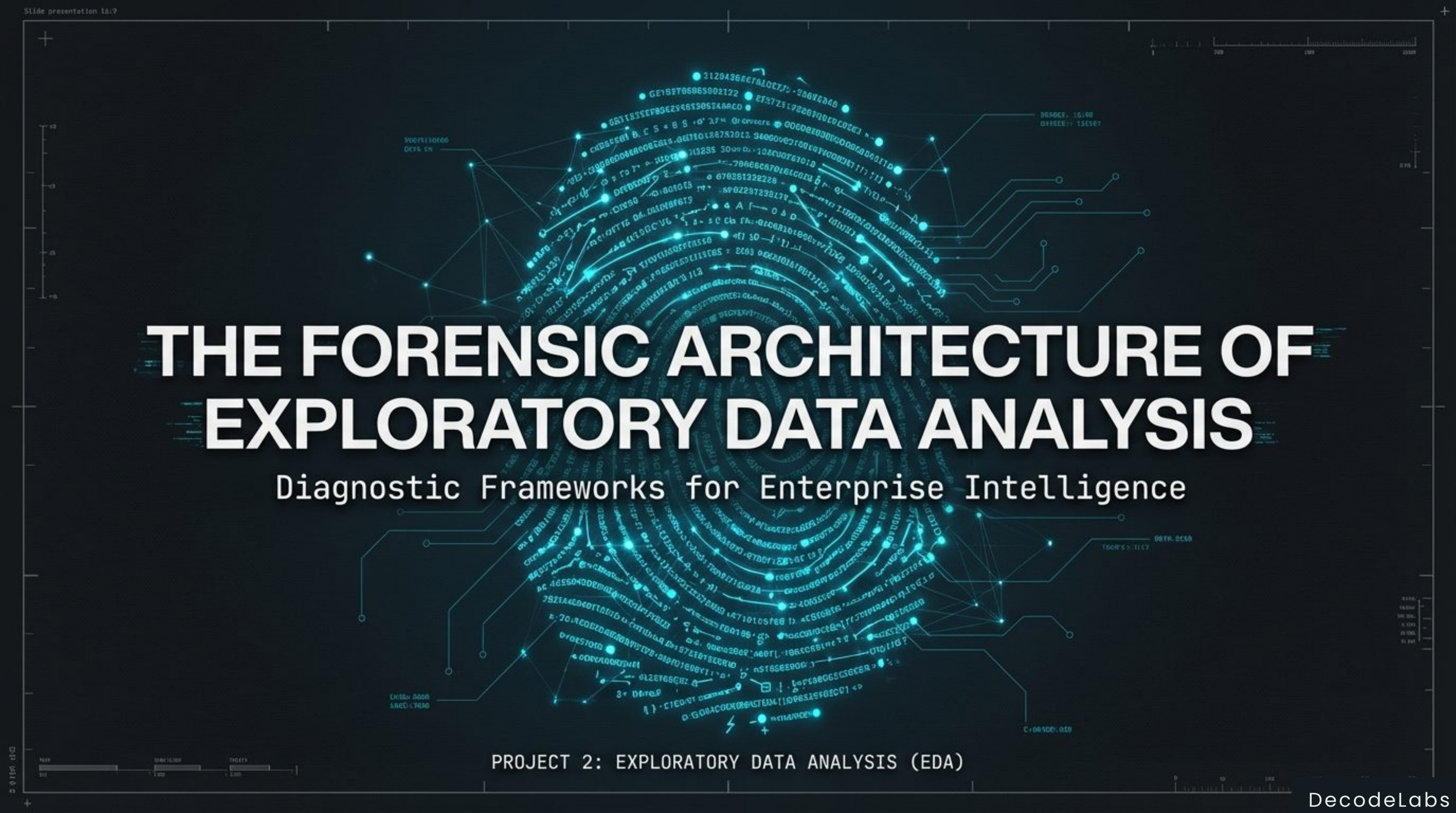
Analyze a dataset to understand patterns, trends, and distributions.

Key Requirements:

- Calculate basic statistics (mean, median, count)
- Identify trends and outliers
- Summarize key observations

Key Skills:

Data analysis, descriptive statistics, analytical thinking


THE FORENSIC ARCHITECTURE OF EXPLORATORY DATA ANALYSIS
Diagnostic Frameworks for Enterprise Intelligence
PROJECT 2: EXPLORATORY DATA ANALYSIS (EDA)

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A Dataset is a Repository of Digital Evidence

THE RAW EVIDENCE

```
ID,Timestamp,Location   ,Value ,Status
001,2023-10-27T10:00:00Z,New York,1200,Valid
002,2023-10-27T10:05:00Z,__,__,Alert
003,INVALID_DATE,London,850,Valid
004,2023-10-27T10:10:00Z,Tokyo,__,Valid
005,2023-10-27T10:15:00Z,Paris,1500,Error
006,2023-10-27T10:20:00Z,__,__,Valid
007,__,Berlin,900,Valid
008,2023-10-27T10:25:00Z,New York,__,Alert
009,2023-10-27T10:30:00Z,Sydney,1100,Valid
```

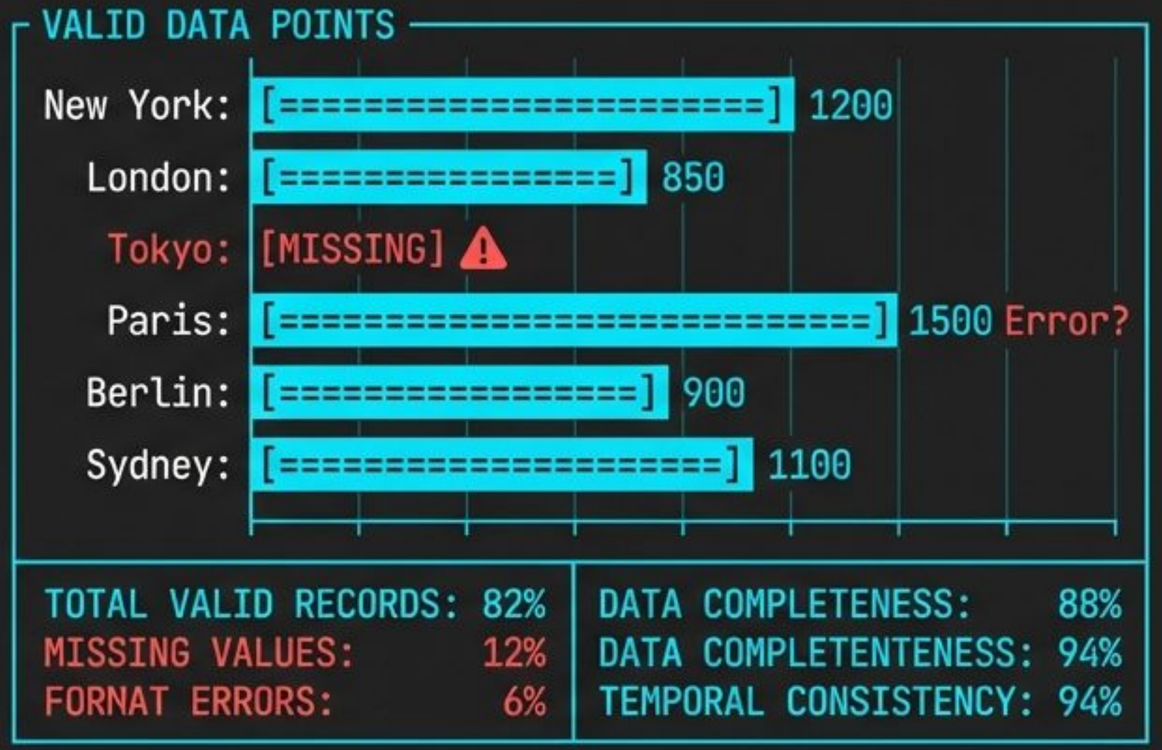
MISSING DATA

INVALID FORMAT

MISALIGNED

DATA CORRUPTION

THE DIAGNOSIS



TOTAL VALID RECORDS:	82%
MISSING VALUES:	12%
FORNAT ERRORS:	6%
DATA COMPLETENESS:	88%
DATA COMPLETENTENESS:	94%
TEMPORAL CONSISTENCY:	94%

"The analyst's objective is to uncover the underlying 'pulse' or patterns of this evidence before any predictive modeling or amplification through visualization occurs."

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The IPO Framework: From Raw Numbers to Business Insight



INPUT

The Evidence.
Raw or Sanitized Data (CSVs, Excel).



PROCESS

The Investigation.
Mathematical transformations & statistical logic.

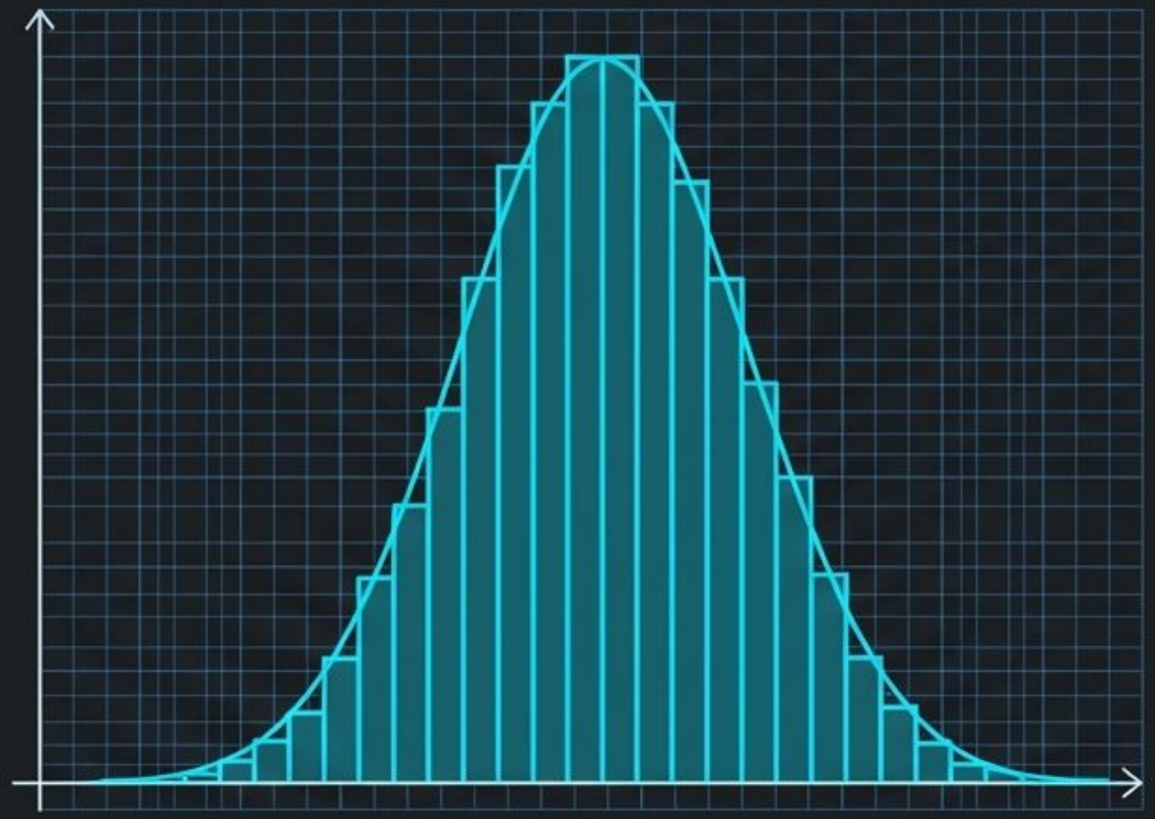


OUTPUT

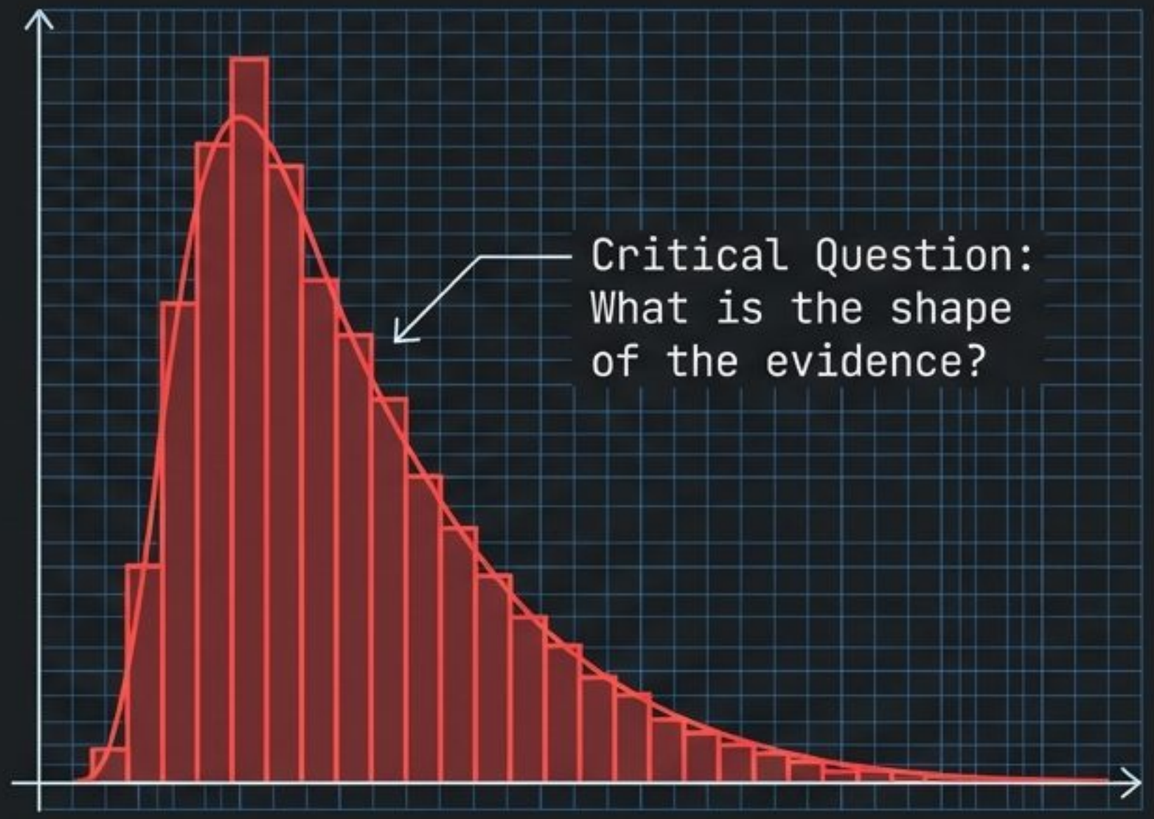
The Verdict.
Actionable insights, summary reports, and decision support.

The Geometry of Distribution

Univariate analysis is the study of one variable’s center of gravity.

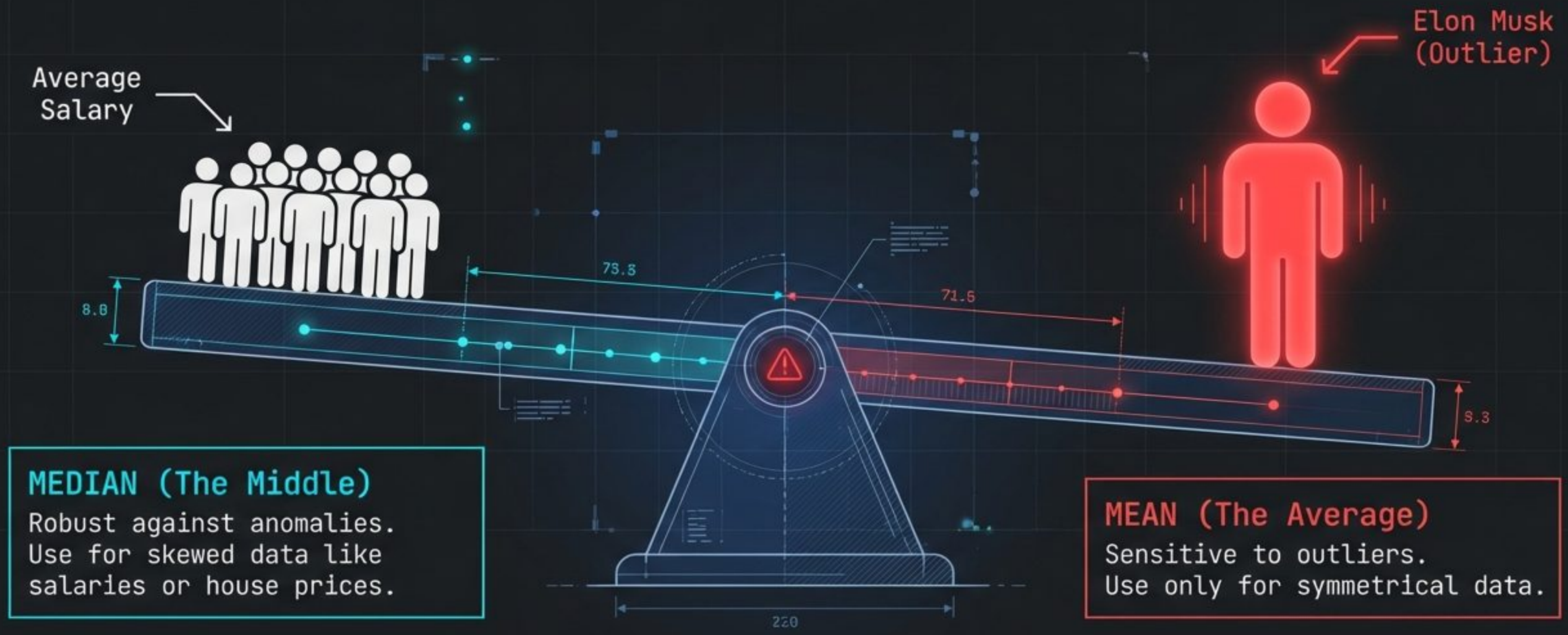


Symmetrical Distribution



Skewed Distribution

Locating the Center of Gravity: Mean vs. Median

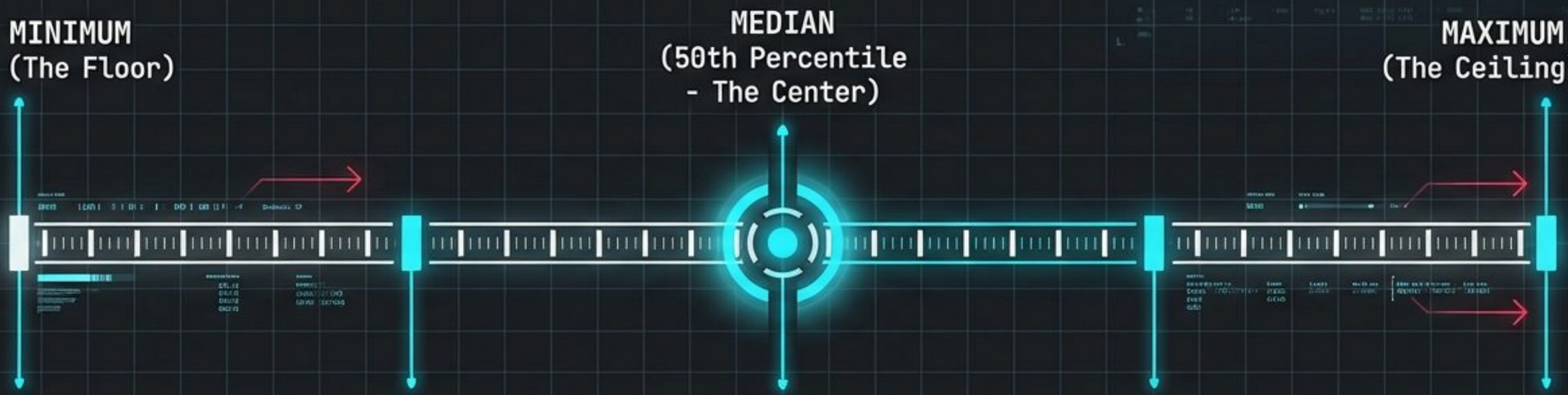


MEDIAN (The Middle)
Robust against anomalies.
Use for skewed data like salaries or house prices.

MEAN (The Average)
Sensitive to outliers.
Use only for symmetrical data.

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The Logic Skeleton: The Five-Number Summary

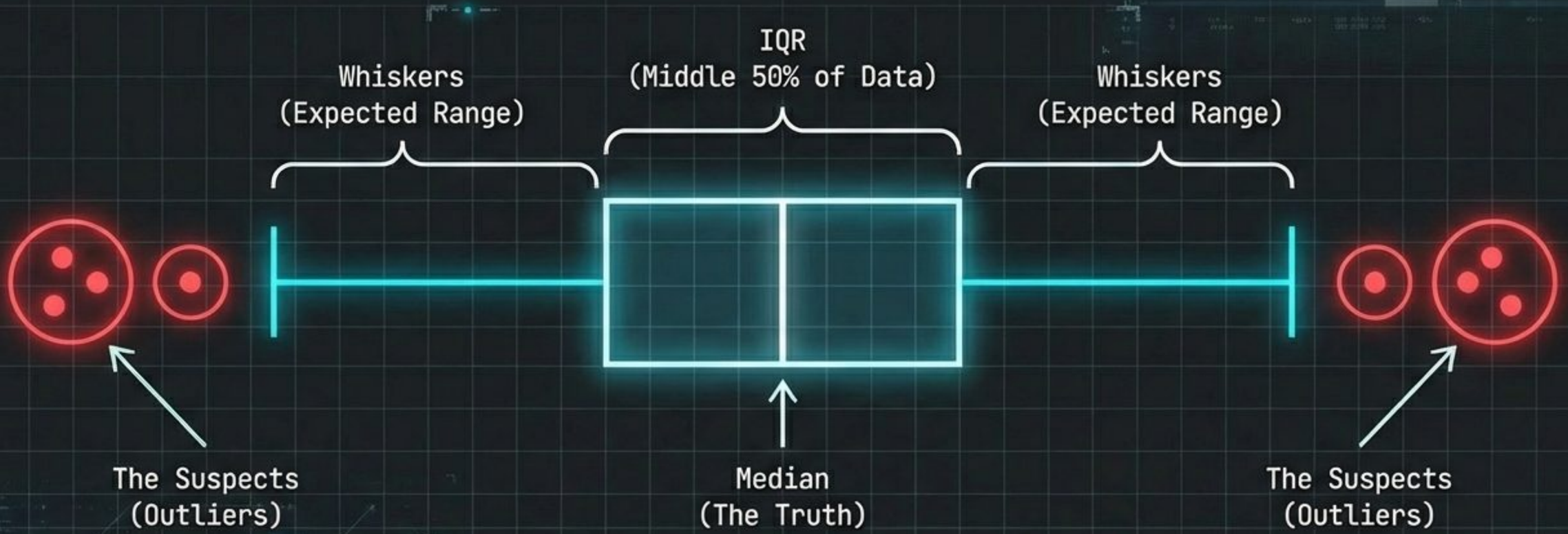



```
Python Tip: df['column'].describe()
```

Unmasking the Outliers: Noise vs. Signal



The Fingerprint of Variability: The Boxplot



Tools of the Trade: IQR vs. Z-Score

The IQR Method



Robust against extreme values.
Best for 'dirty' business data.

$$\text{Outlier} < Q1 - 1.5 \times IQR \quad \text{OR} \quad \text{Outlier} > Q3 + 1.5 \times IQR$$

The Z-Score Method

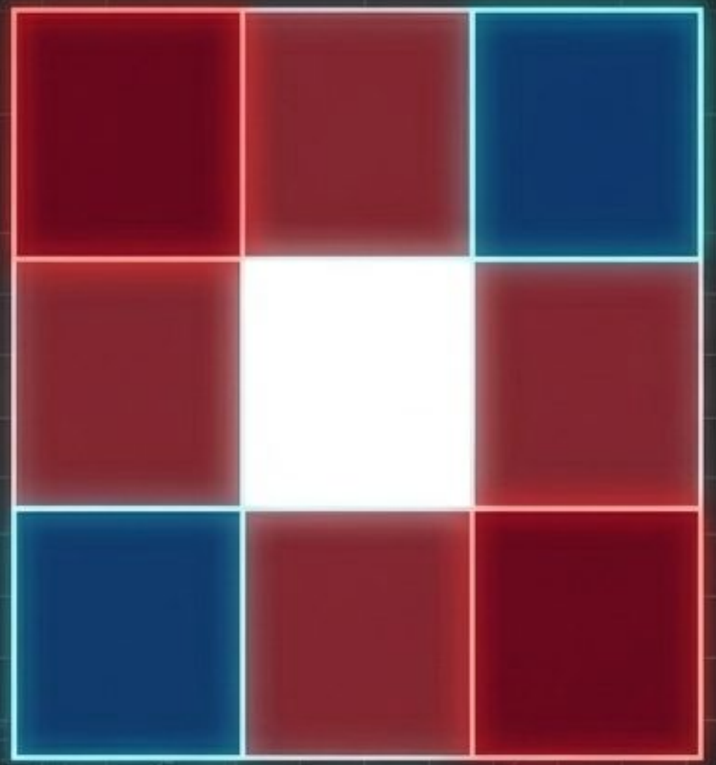


Sensitive to extreme values.
Best for Normal Distributions.

$$Z = \frac{(\text{Value} - \text{Mean})}{\text{Standard Deviation}}$$

Flag if $Z > 3$ or $Z < -3$

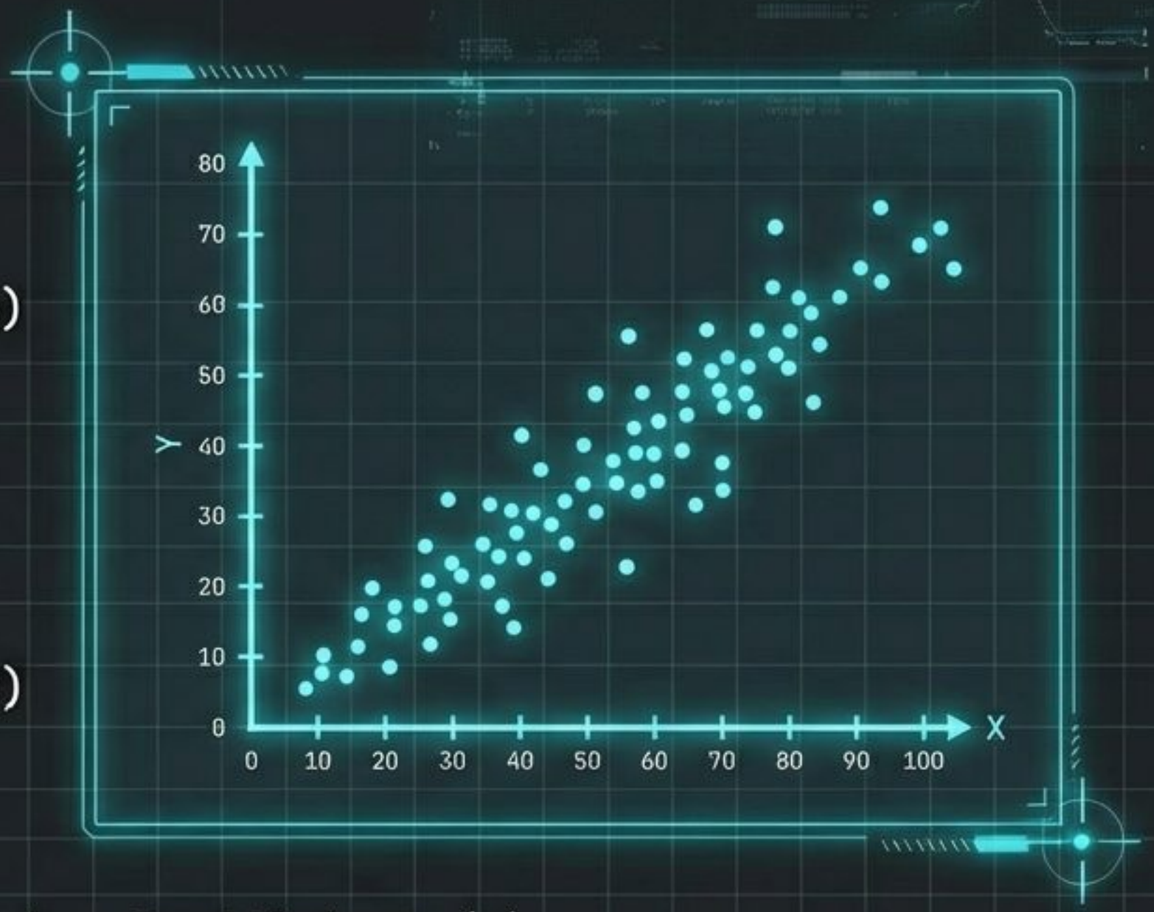
Mapping Relationships: Correlation Analysis



Dark Red = +1.0
(Perfect Positive)

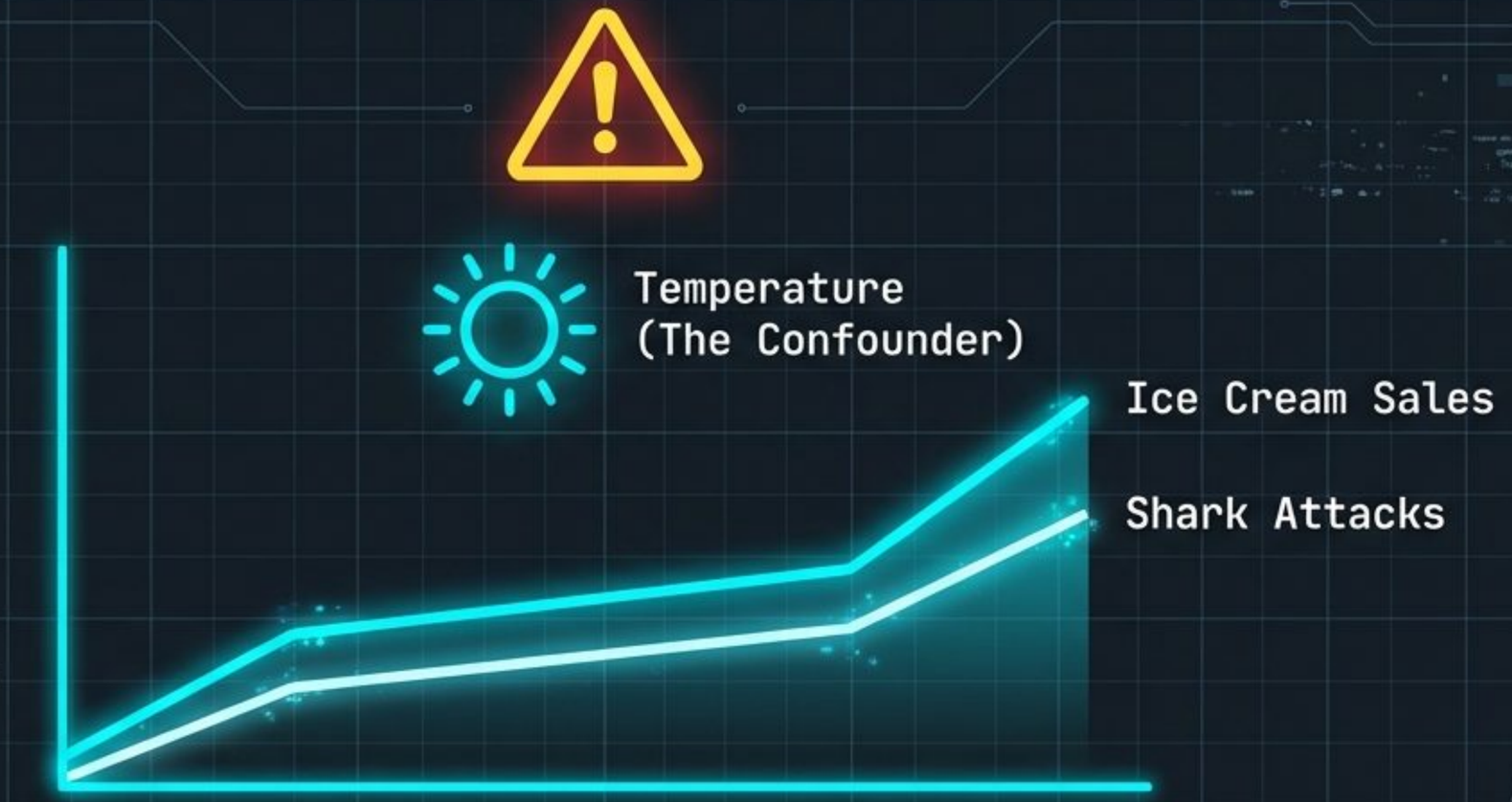
White = 0.0
(No Correlation)

Dark Blue = -1.0
(Perfect Negative)



Metric: Pearson Correlation Coefficient (r).
Used to measure strength and direction of linear relationships.

The Golden Rule: Correlation ≠ Causation



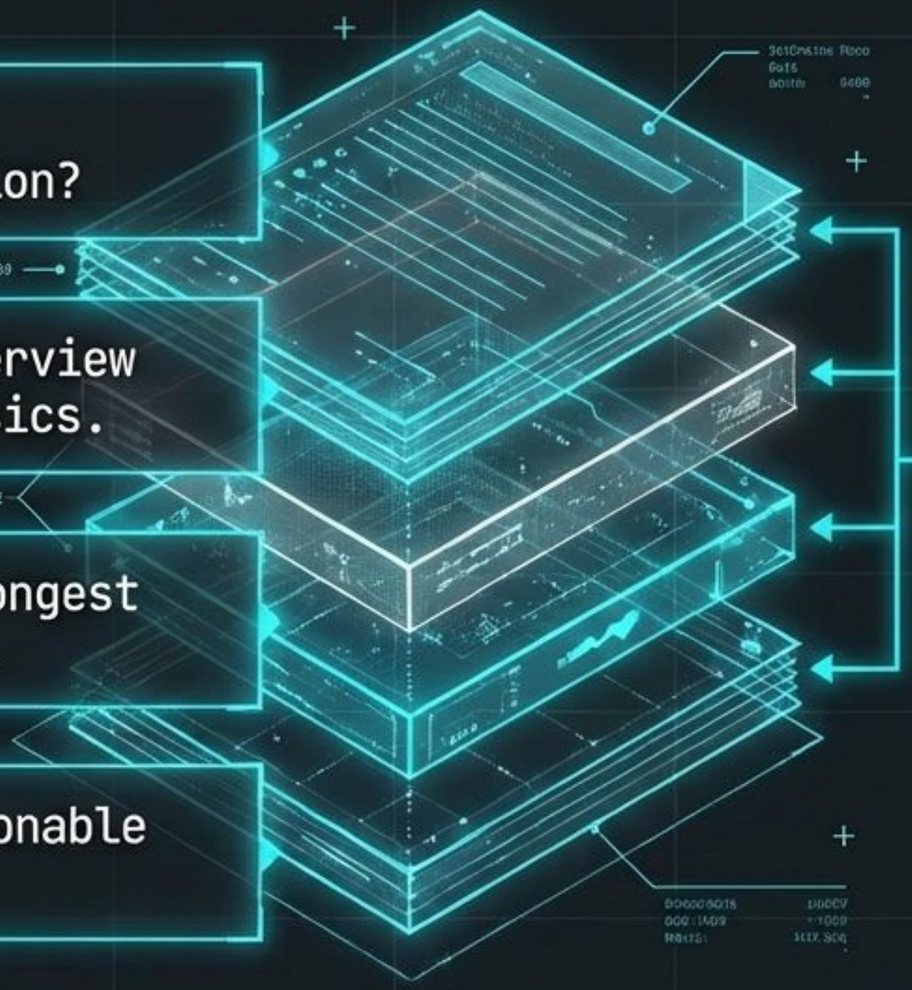
Forensic Note: A correlation is a CLUE, not a VERDICT.
Always check for hidden variables.

Synthesizing the Case: The “So What?” Test

WHAT THE DATA SAYS		THE BUSINESS DIAGNOSIS
Mean churn increased by 3%.	Translate to Impact	We lost 1,510 customers, equivalent to closing 11 retail stores.
Income data is right-skewed.		Standard marketing will fail for 70% of our base; we need a low-cost segment strategy.

The Executive Summary Architecture

1. **Problem Statement:** What is the core question?
2. **Methodology:** Brief overview of data cleaning/forensics.
3. **Key Findings:** The strongest signals and predictors.
4. **Recommendations:** Actionable next steps.



Pro Tip: Headlines First. Don't title a slide "Data". Title it "Revenue Grew 19%".

Visual Evidence: Designing for Stakeholders

Before and After

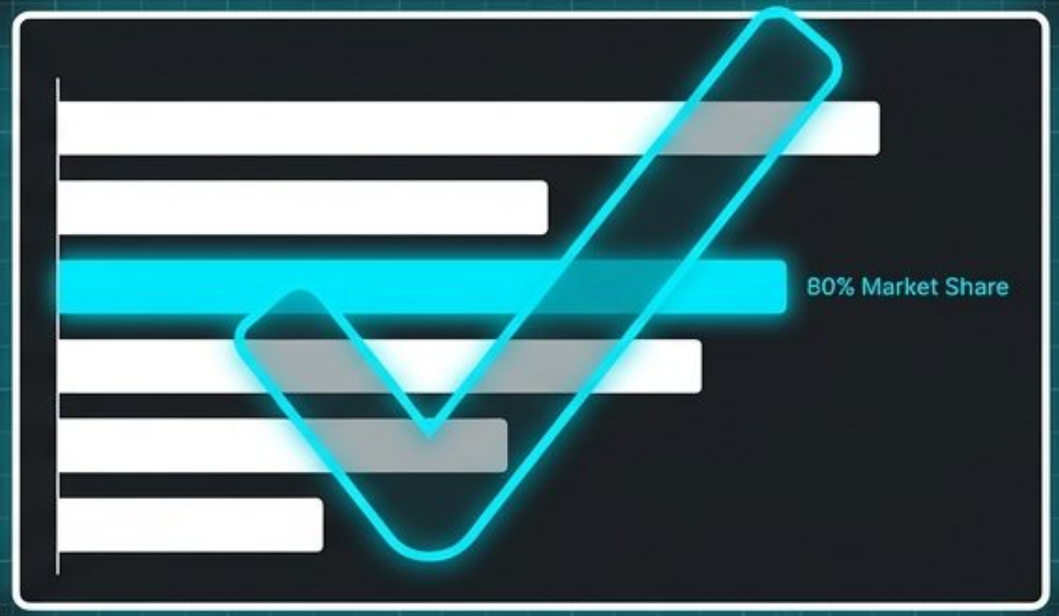
The Noise

Clinical White JetBrains Mono



The Insight

Clinical White JetBrains Mono



Best Practices:

- Cut the noise.
- Highlight the insight.
- Remove 3D effects.

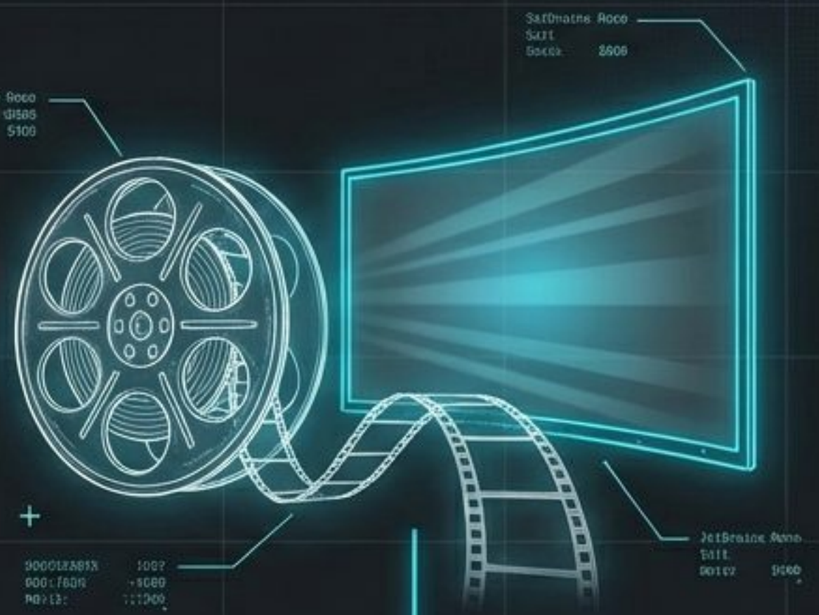
Portfolio Architecture: Proving Your Expertise



Resume
(The Trailer).

87%

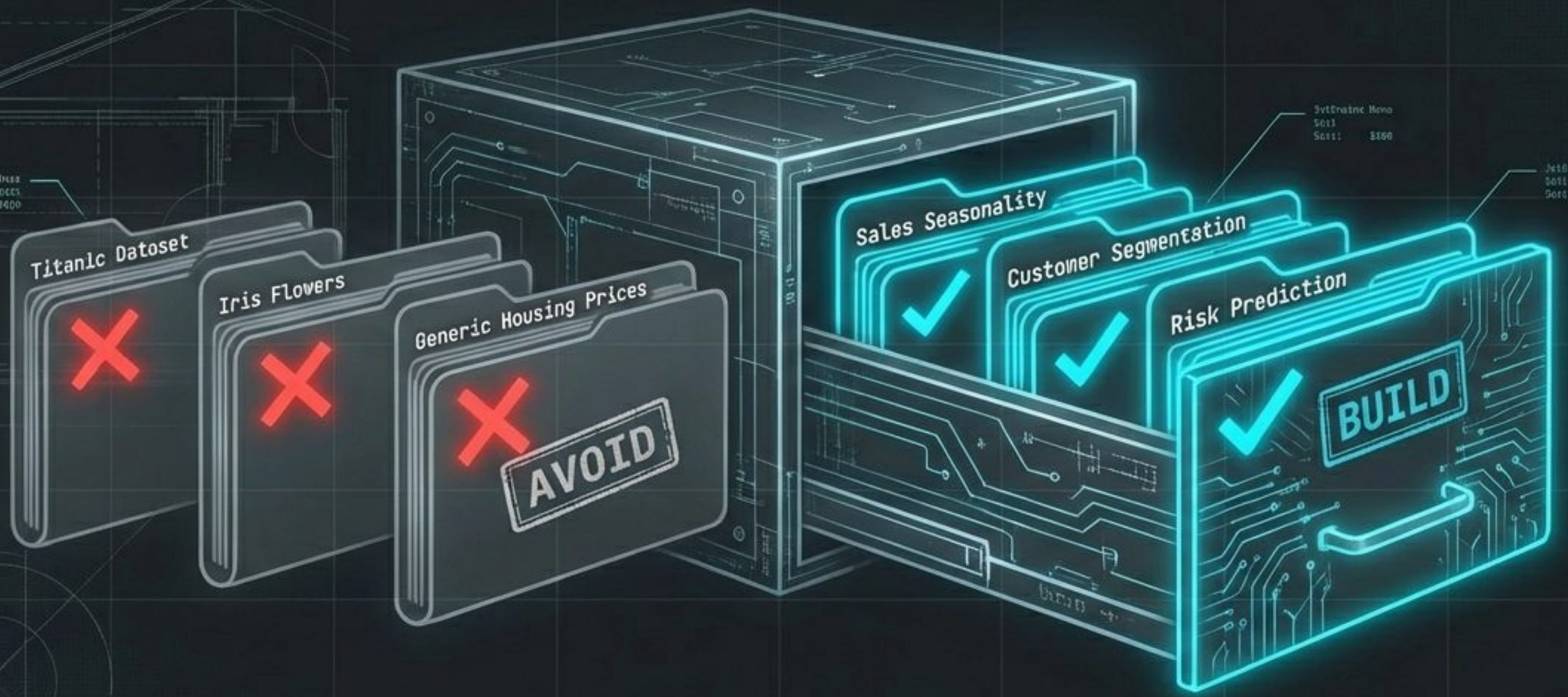
of hiring managers are more likely to interview candidates with portfolios.



Portfolio
(The Feature Film).

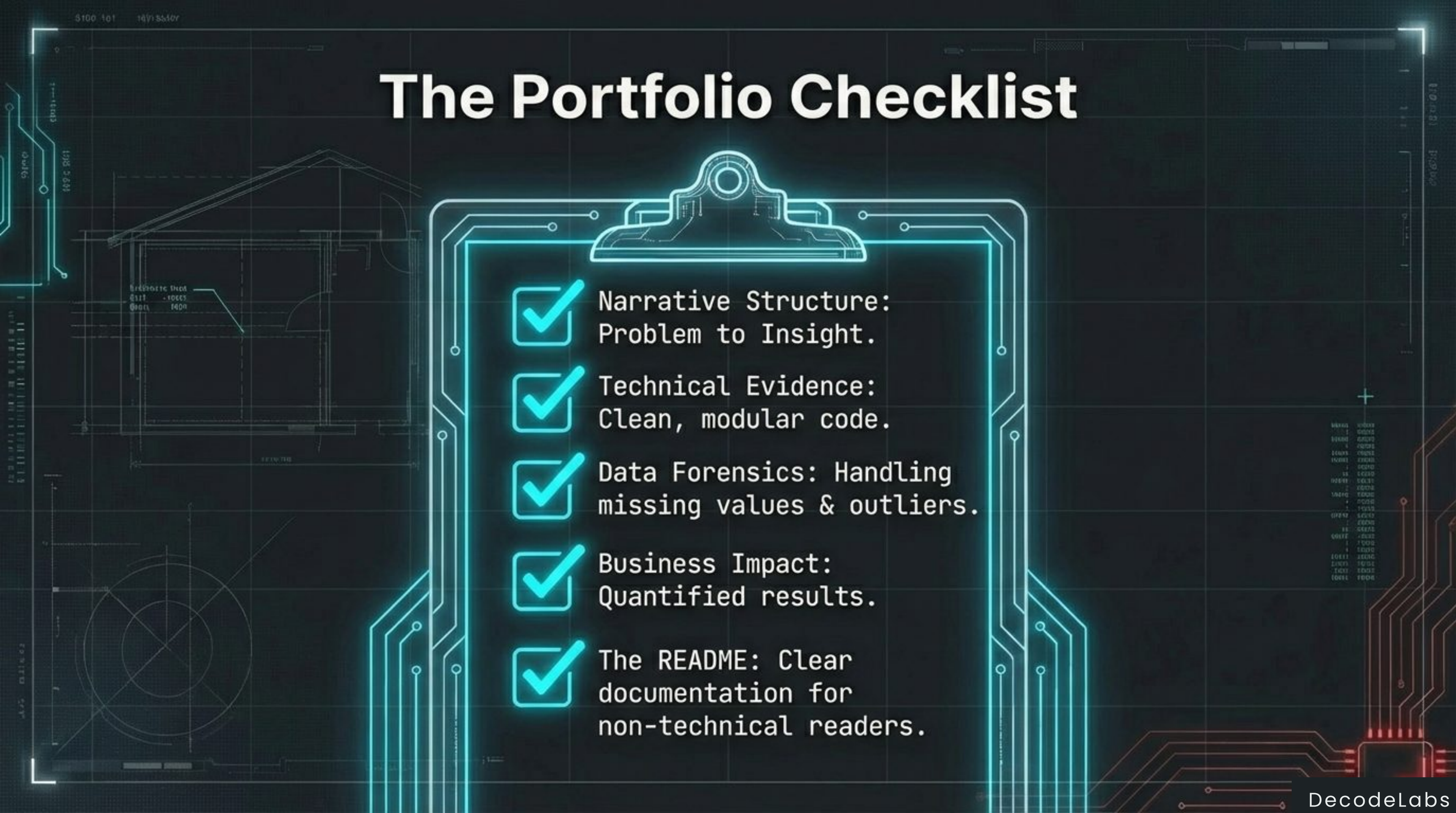
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Selecting Your Case Files: Project Strategy

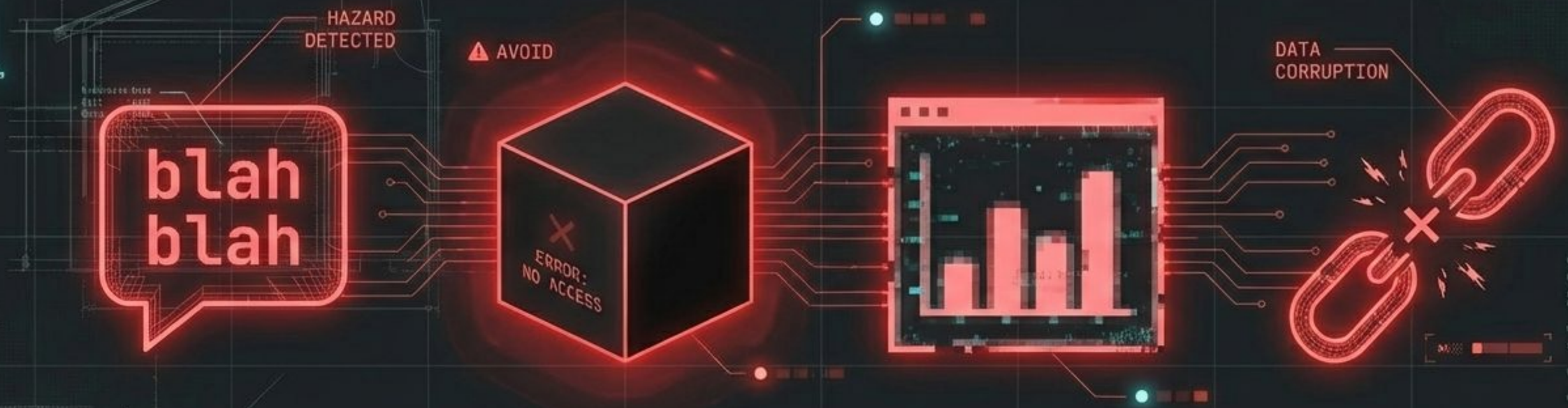


Don’t use ‘Hello World’ datasets.
Build unique, business-relevant projects.

The Portfolio Checklist



Common Mistakes to Avoid



Jargon Overload.

The Black Box (No methodology).

Static Charts (Screenshots of code).

Ignoring Context.

The Verdict



- EDA is the diagnostic phase for data integrity.
- Translate Noise into Signal.
- Uncover the Truth.

“You are the translator between data and decision.”

CONCLUSION

The absolute best way to master Data Analytics is through hands-on practice, not just theory. Don't just aim to complete these projects; take them one by one, experiment with unique solutions—like exploring correlations between variables or visualizing distributions—and treat every unexpected outlier as a learning opportunity. As you build these milestones at DecodeLabs, you are creating a real-world portfolio that showcases your analytical depth to future employers. Your journey to becoming a professional analyst begins right here, right now, with the very first pattern you discover today.

THANK YOU

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